IdiomX: A Bilingual Benchmark for Idiom Understanding, Generation, and Cross-Lingual Retrieval

Ayman Ali Sharara
MSc Data Science & Machine Learning (SPOC S21)
Data ScienceTech Institute (DSTI)
Email: ayman.sharara@edu.dsti.institute

Supervisor: Hanna Abi Akl
Email: hanna.abi-akl@dsti.institute

Abstract

Idiomatic expressions remain a major challenge for natural language processing because their meanings are often non-literal, context-dependent, and difficult to align across languages. In this work, we introduce IdiomX, a large-scale bilingual benchmark for idiom understanding built through a multi-stage pipeline combining lexical-resource collection, rule-based filtering, LLM-based enrichment, and validation. The resulting dataset contains 123,336 contextualized examples covering 14,986 English idioms. Each idiom is associated with multiple generated contexts, including both idiomatic and literal usages, enabling balanced evaluation of compositional and non-compositional language understanding.

IdiomX is constructed from publicly usable lexical resources, primarily WordNet and Kaikki.org, followed by normalization and idiom-focused filtering to retain globally meaningful expressions.

We then use OpenAI GPT-4.1-mini in a controlled structured-generation pipeline to enrich each idiom with contextual examples, canonical and surface-form information, English and Arabic meanings, translations, and additional semantic metadata. The enrichment stage is followed by schema validation, consistency checks, lexical alignment checks, anomaly filtering, and targeted manual inspection.

We benchmark IdiomX on three tasks: idiom detection, context-to-idiom prediction, and Arabic context-to-idiom retrieval. Experiments include classical baselines, transformer-based classifiers, dense retrieval models, hybrid generation–retrieval systems, and task-specific contrastive fine-tuning. Results show that transformer models substantially outperform lexical baselines in idiom detection, the proposed hybrid model achieves the strongest performance for context-to-idiom prediction, and fine-tuned multilingual E5 retrieval produces the best Task 3 results, reaching 64.86% Top-1 and 81.46% Top-5 accuracy in the upgraded cross-lingual retrieval setting.

These findings demonstrate that idiom understanding benefits from combining high-quality bilingual supervision with task-specific modeling. IdiomX provides a strong foundation for future work in figurative language understanding, multilingual semantic retrieval, and idiom-aware language modeling.

Keywords: idioms; figurative language; bilingual benchmark; cross-lingual retrieval; Arabic NLP; large language models

I. Introduction

Idiomatic expressions are a fundamental and challenging component of natural language, as their meanings are often non-compositional and cannot be directly inferred from the meanings of their individual words. This makes idioms particularly difficult for natural language processing systems across tasks such as semantic understanding, translation, retrieval, and generation.

For example, the phrase “kick the bucket” cannot be interpreted compositionally, while even more transparent idioms may still require strong contextual reasoning to distinguish idiomatic from literal usage, as shown in prior work on idiom detection and cross-lingual semantic modeling.

Despite their importance, existing idiom resources remain limited in one or more dimensions, including scale, contextual richness, annotation depth, multilingual coverage, and benchmarking support.

Many datasets contain only a small number of idiom types (e.g., verb–noun constructions, fixed expressions, or phrasal idioms), focus on a single language, or provide narrow annotations that are insufficient for modern deep learning and cross-lingual modeling.

To address these limitations, we present IdiomX, a large-scale bilingual benchmark for idiom understanding and modeling. IdiomX combines lexical-resource collection with controlled large language model (LLM) enrichment to produce contextualized idiom data with English–Arabic semantic alignment. The dataset is designed not only as a resource, but as a unified benchmark supporting multiple tasks that reflect different aspects of idiom understanding.

Each idiom in IdiomX is associated with multiple contextual examples covering both idiomatic and literal usages. This paired design enables precise evaluation of models' ability to distinguish figurative meaning from compositional interpretation, which remains a core challenge in figurative language processing.

A key motivation of this work is that idiom understanding should be studied across both monolingual and multilingual settings. In particular, mapping Arabic contextual descriptions to canonical English idioms defines a challenging retrieval task that emphasizes semantic alignment rather than surface similarity. This makes IdiomX suitable for evaluating both contextual language models and retrieval-based systems under realistic ambiguity.

The main contributions of this paper are as follows:

1. We introduce IdiomX, a large-scale bilingual idiom benchmark containing 123,336 contextualized examples (balanced between idiomatic and literal usage) covering 14,986 idioms.
2. We propose a scalable data construction pipeline combining lexical-resource collection, LLM-based enrichment, structured validation, and targeted manual inspection.
3. We provide rich English-Arabic aligned semantic fields, enabling cross-lingual modeling and retrieval.
4. We benchmark IdiomX on three tasks: idiom detection, context-to-idiom prediction, and Arabic context-to-idiom retrieval.
5. We demonstrate that hybrid modeling improves context-to-idiom prediction, while task-specific contrastive fine-tuning significantly enhances cross-lingual retrieval performance, achieving up to 64.86% Top-1 and 81.46% Top-5 accuracy in Task 3.

Overall, IdiomX provides both a reusable dataset and a comprehensive benchmark for studying idiomatic language across modeling paradigms and languages.

II. Related Work

A. *Idiom Datasets*

Idiomatic expression understanding has been widely studied across several NLP tasks, including idiom detection, interpretation, and paraphrasing. Early work focused on lexical resources such as WordNet and Wiktionary, which provide structured definitions and semantic relations but lack contextual diversity and large-scale annotations.

For example, in the VNC-Tokens dataset, the phrase "take place" may be labeled as idiomatic when meaning "to occur", but literal in contexts describing physical movement. Similarly, SemEval-2013 Task 5b includes expressions such as "make up", which can denote either "to invent" or "to compose", depending on context.

Several benchmark datasets have been introduced for idiom detection and classification. Notable examples include VNC-Tokens and SemEval-2013 Task 5b (e.g., VNC-Tokens; SemEval-2013 Task 5b), which provide labeled instances of verb–noun constructions and idiomatic expressions in context. However, these datasets are limited in size and do not support modern deep learning requirements.

More recent datasets, such as MAGPIE and PIE-English, increase scale and contextual coverage. Nevertheless, they remain monolingual and typically lack rich semantic annotations or multilingual alignment, limiting their applicability for cross-lingual modeling.

B. *Neural Models for Idiom Understanding*

Recent work has explored neural approaches to idiom understanding using contextual language models and sequence-to-sequence architectures. Transformer-based models such as BERT have been widely used for idiom detection and classification, where the objective is to distinguish figurative from literal usage in context. Sequence-to-sequence models such as T5 have also been applied to idiom interpretation, paraphrasing, and generation tasks, benefiting from their ability to model semantic reformulation.

For instance, transformer-based models such as BERT classify sentences like "He kicked the bucket" as idiomatic, while sequence-to-sequence models such as T5 can generate paraphrases such as "He passed away". These examples highlight the importance of contextual modeling in idiom understanding.

These approaches have achieved strong results when sufficient contextual supervision is available. However, their effectiveness remains closely tied to dataset quality, scale, and diversity. Cross-lingual idiom understanding remains much less studied than monolingual detection or interpretation, and retrieval-oriented settings are still underrepresented.

This motivates the need for a benchmark that supports not only idiom classification, but also generation and cross-lingual retrieval under realistic contextual ambiguity.

To better position IdiomX within this landscape, Table 1 provides a quantitative comparison with widely used idiom datasets.

Dataset	Lang	Idioms	Inst.	Bi	Ref
VNC-Tokens	EN	53	2,984	No	ACL Anthology
SemEval-2013 Task 5b	EN	85	~4,350	No	ACL Anthology
IDIX	EN	52	4,022	No	ACL Anthology
MAGPIE	EN	~1,700	56,000	No	ACL Anthology
PIE-English	EN	~1,200	~20,100	No	arXiv
IdiomX (Ours)	EN+AR	14,986	123,336	Yes	GitHub[1]

Table 1: Comparison of IdiomX with existing idiom datasets

[1] Dataset repository link:
https://github.com/aymanshar/idiomx-dataset
https://github.com/aymanshar/IdiomX

This comparison highlights the gap between existing monolingual datasets and the need for large-scale bilingual resources with structured semantic annotations.

Beyond scale, IdiomX distinguishes itself through its balanced construction of idiomatic and literal examples, as well as its rich linguistic annotations, enabling deeper analysis of ambiguity, compositionality, and semantic consistency. In addition, IdiomX integrates controlled LLM-based enrichment to generate diverse contextual examples and bilingual semantic representations, supporting a broader range of tasks including detection, generation, and cross-lingual retrieval.

III. DATASET CONSTRUCTION

The IdiomX dataset is constructed through a multi-stage pipeline combining data collection, preprocessing, enrichment, and validation.

First, idiomatic expressions are collected from multiple lexical sources, including WordNet and Kaikki.org (a structured Wiktionary-derived resource), selected due to their open licensing and suitability for public dataset release. Duplicate entries and low-quality records (e.g., incomplete entries, malformed expressions, or non-idiomatic multiword phrases) are removed through normalization and filtering.

Next, contextual examples are generated and enriched using a large language model (LLM). For each idiom, multiple example sentences are produced, covering both idiomatic and literal usage. The enrichment process also generates English and Arabic meanings, as well as aligned translations of the example sentences.

For example, for the idiom "kick the bucket", the system generates an idiomatic sentence such as "He finally kicked the bucket after a long illness", alongside a literal variant such as "He kicked the bucket across the yard", together with aligned English and Arabic meanings.

This paired generation setup allows the same surface form to appear in both figurative and literal contexts, which is essential for controlled evaluation.

To ensure consistency, additional linguistic features are computed, including compositionality level, learner difficulty, and semantic similarity scores. These features provide deeper insights into idiomatic behavior and support advanced modeling tasks.

The final dataset consists of over 123,000 examples with rich annotations and bilingual coverage.

The overall dataset construction pipeline is illustrated in Figure 1.

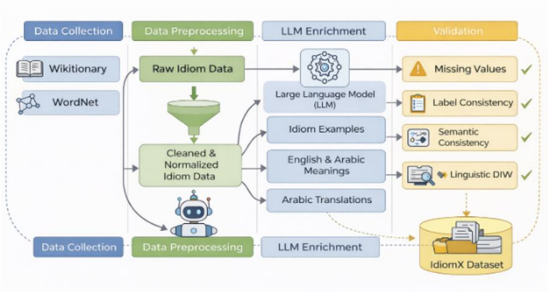

Figure 1: Overview of the IdiomX dataset construction pipeline

A. *Data Sources and Collection*

Idiomatic expressions are collected from publicly available lexical resources, primarily WordNet and Kaikki.org (a structured Wiktionary-derived resource). These sources were selected due to their open licensing, structured format, and suitability for building a reproducible and publicly shareable dataset.

Initial extraction focuses on multiword expressions and candidate idiomatic phrases. Duplicate entries and non-idiomatic expressions are removed through normalization and rule-based filtering, resulting in a high-precision set of canonical idioms.

B. *Filtering and Normalization*

The collected expressions undergo a strict filtering process to ensure idiomatic validity and consistency. This includes removal of compositional phrases, normalization of surface forms, and elimination of duplicates across sources.

Additional heuristics are applied to retain globally meaningful idioms while filtering out rare, domain-specific, or noisy expressions. This step ensures that the resulting dataset focuses on widely used idiomatic language suitable for general NLP applications.

This filtering step prioritizes precision over recall to ensure that the resulting dataset contains high-confidence idiomatic expressions suitable for downstream modeling.

C. *Example Generation Strategy*

To ensure sufficient contextual diversity and enable controlled evaluation, each idiom in the dataset is associated with multiple generated examples covering both idiomatic and literal usages.

Specifically, for each idiom, four idiomatic examples are generated to reflect its figurative meaning, and four literal examples are constructed using the same lexical components but without idiomatic interpretation. This results in approximately eight contextual instances per idiom.

This design produces a balanced dataset with 61,695 idiomatic examples and 61,641 literal examples. The balanced structure is particularly important for idiom detection tasks, where models must distinguish between figurative and compositional usage under similar surface forms.

D. *LLM-based enrichment and Prompting Strategy*

Following data collection and filtering, we apply a large language model (LLM)-based enrichment process to generate structured semantic annotations and contextual examples.

We use OpenAI GPT-4.1-mini in a controlled batch inference pipeline with structured JSON outputs. This model is selected as a trade-off between generation quality and scalability, enabling large-scale data enrichment while maintaining consistent output quality.

The model is prompted to generate multiple fields per idiom, including:

- Contextual examples.
- Idiomatic and literal usage labels.

- Canonical and surface forms.
- English and Arabic meanings.
- Semantic consistency indicators.
- Compositionality levels and learner difficulty estimates.

Prompting is carefully designed to enforce consistency across outputs and to ensure that generated examples adhere strictly to idiomatic or literal usage as required. The use of structured JSON outputs enables reliable parsing, validation, and downstream processing.

To illustrate the prompting strategy, the model is given structured instructions to generate outputs in JSON format. For example, a prompt specifies the idiom, required number of idiomatic and literal examples, and expected fields such as usage label, meaning, and translations. The output follows a fixed schema including fields such as "idiom_surface", "example", "usage_label", and "meaning_en".

Batch processing is used to scale the enrichment pipeline efficiently, allowing generation over tens of thousands of idioms while maintaining reproducibility and consistency across runs.

For instance, a generated record may include fields such as {"idiom_surface": "kick the bucket", "usage_label": "idiomatic", "meaning_en": "to die", "example": "He finally kicked the bucket after a long illness"}. This fixed schema enables reliable automatic validation and downstream task construction.

E. Data Validation and Quality Control

To ensure data quality, a multi-stage validation pipeline is applied to the enriched dataset. This includes schema validation, consistency checks, anomaly detection, and targeted manual inspection of suspicious records.

Semantic consistency is measured using cosine similarity between embedding representations of idioms and their contextual sentences.

For example, strict matching requires exact string alignment between the idiom and its occurrence in the sentence, while relaxed matching allows minor variations such as inflectional changes or word reordering.

Each record is assigned a validation status, resulting in:

- 121,079 valid entries
- 1,531 verified entries
- 726 corrected entries

Additional semantic metadata is analyzed to assess dataset properties. For example, idiom compositionality is distributed across semi-opaque (71,296), opaque (29,472), and transparent (22,568) categories. Ambiguity levels indicate that 80,904 examples are semi-literal possible, while 30,720 are strongly idiomatic.

Learner difficulty annotations show that most idioms fall within the medium difficulty range (89,528), with smaller proportions classified as easy (17,552) and hard (16,256).

These statistics confirm that the dataset captures a diverse range of idiomatic behavior, ambiguity levels, and semantic complexity, making it suitable for robust evaluation across multiple tasks.

This multi-stage validation process ensures that the dataset maintains high structural consistency and semantic reliability, which is critical for training and evaluating deep learning models.

F. Dataset Analysis and Distribution

To better understand the characteristics of the IdiomX dataset, we analyze the distribution of key linguistic and structural features, focusing on label balance, compositionality, ambiguity, and semantic quality.

As shown in Figure 2 illustrates the distribution of idiomatic versus literal examples, confirming the effectiveness of the controlled generation strategy and ensuring a near-perfect class balance.

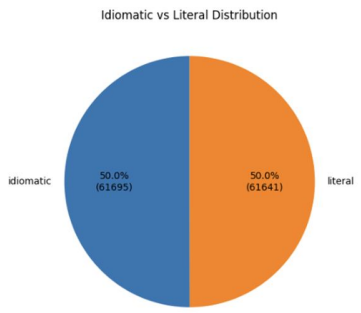

Figure 2: Idiomatic vs Literal Distribution

We further analyze key linguistic attributes of the dataset, including ambiguity and compositionality, as shown in Figures 3 and 4.

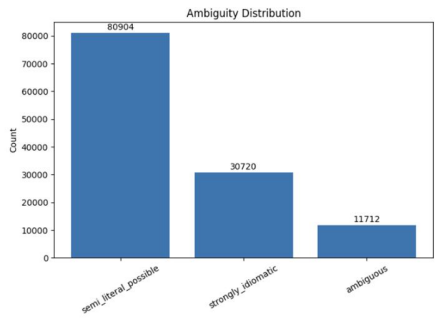

Figure 3: Ambiguity Distribution

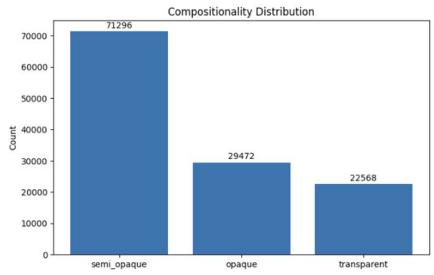

Figure 4: Compositionality Distribution

As shown, the dataset exhibits a wide range of ambiguity levels, with a large proportion of semi-literal cases, reflecting

realistic linguistic scenarios where idiomatic meaning may overlap with compositional interpretation. Similarly, the compositionality distribution spans semi-opaque, opaque, and transparent idioms, indicating that the dataset captures varying degrees of semantic complexity.

For example, partially compositional idioms such as "spill the beans" retain some connection to their literal meaning while still requiring contextual interpretation.

Figures 5 and 6 present the distributions of example length, semantic consistency, and canonical semantic scores.

The semantic consistency and canonical similarity distributions exhibit well-formed distributions that approximate Gaussian behavior.

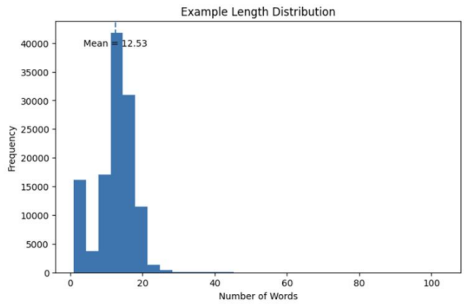

Figure 5: Example Length Distribution

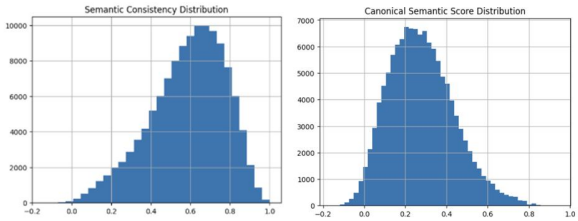

Figure 6: Semantic Consistency and Canonical Semantic Distribution

Finally, Figures 7, 8, and 9 present deeper semantic quality analysis across labels, difficulty levels, and domains.

As shown in Figure 7 shows that idiomatic examples exhibit lower semantic similarity to their literal forms but maintain strong internal consistency, confirming that the dataset effectively distinguishes between figurative and compositional usage.

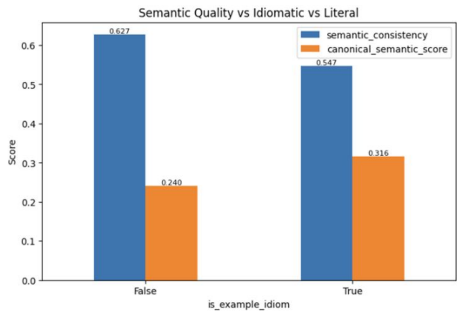

Figure 7: Semantic Quality vs Label

As shown in Figure 8 highlights variation across learner difficulty levels, where more difficult idioms tend to exhibit lower semantic similarity, reflecting increased abstraction and reduced compositional transparency.

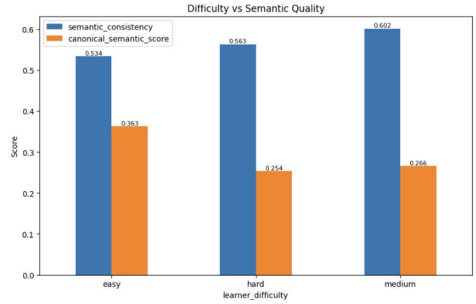

Figure 8: Difficulty vs Semantic Quality

As shown in Figure 9 illustrates the domain distribution, showing that while the dataset is dominated by general-domain idioms, it also includes a diverse range of specialized domains, supporting broader applicability across NLP tasks.

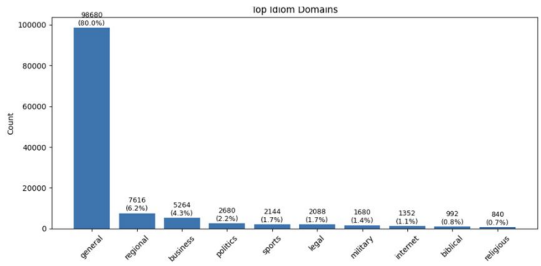

Figure 9: Domain Distribution

IV. Benchmark Tasks

IdiomX supports multiple downstream NLP tasks involving idiomatic language.

A. *Task 1: Idiom Detection.*

B. *Task 2: Context-to-Idiom Prediction.*

C. *Task 3: Arabic Context-to-Idiom Retrieval.*

Similar tasks have been explored in prior work on idiom understanding and figurative language processing.

Idiom detection has been studied as a classification problem (e.g., SemEval-2013 Task 5b), while context-to-idiom prediction relates to lexical substitution and paraphrase generation tasks.

Cross-lingual idiom retrieval remains less explored but is related to multilingual semantic retrieval and cross-lingual sentence matching using transformer-based models.

These benchmark tasks are made possible by the structured design of IdiomX. Specifically, the dataset includes fields such as:

- The input sentence (idiom_in_example),
- binary usage labels (is_example_idiom),

And aligned semantic representations in both English and Arabic:

- o idiom_in_example_meaning_en,
- o idiom_in_example_meaning_arabic.

These fields were explicitly introduced during dataset construction to enable supervised learning, evaluation, and cross-lingual benchmarking across multiple idiom understanding tasks.

V. EXPERIMENTS

To evaluate the effectiveness of IdiomX, we conduct baseline experiments across multiple tasks supported by the dataset.

A. *Task 1: Idiom Detection*

Idiom detection is formulated as a binary classification task, where the objective is to determine whether a given sentence expresses an idiomatic meaning or a literal interpretation.

This task plays a critical role in natural language understanding, as idiomatic expressions often convey meanings that cannot be inferred directly from individual words. Accurate detection is essential for applications such as:

- Machine translation,
- Search systems,
- Conversational agents,
- Large language model (LLM) evaluation, and
- Language Learning Tools.

Unlike standard text classification, idiom detection requires distinguishing between surface-level lexical patterns and deeper contextual meaning, making it a challenging semantic task.

1. *Pipeline Overview*

The idiom detection pipeline consists of three main stages:

(1) Data Preparation, preprocessing and split construction

The dataset is constructed from the IdiomX resource, where each instance includes an input sentence and a binary label indicating literal or idiomatic usage. The data is cleaned by removing invalid entries, filtering labels, and eliminating duplicates. A balanced splitting strategy is applied to ensure fair evaluation across training, validation, and test sets.

(2) Feature Representation

Different feature representations are used depending on the model type. For classical models, TF–IDF features with uni-grams and bi-grams are extracted to capture surface-level lexical patterns. For transformer-based models, tokenized sentences are processed using pretrained language models to obtain contextual embeddings.

(3) Model training and Evaluation

Models are trained using supervised learning. Logistic Regression is trained on TF–IDF features, while RoBERTa is fine-tuned end-to-end using cross-entropy loss. All experiments are conducted under consistent preprocessing and evaluation settings to ensure reproducibility.

2. *Models*

We formulate idiom detection as a binary classification task. Given an input sentence $x = (w_1, w_2, \ldots, w_n)$, where each w_i represents a word token, the model predicts a label $y \in \{0,1\}$, where $y = 1$ indicates idiomatic usage and $y = 0$ indicates literal usage.

We evaluate both classical machine learning models and transformer-based architectures.

For classical models, we use Logistic Regression with TF–IDF features (uni-grams and bi-grams). The model estimates the probability of idiomatic usage as:

$$P(y = 1 \mid x) = \sigma(w^T x + b)$$

Where σ is the sigmoid function. The model is trained using binary cross-entropy loss and primarily captures surface-level lexical patterns.

For transformer-based models, we use RoBERTa, which computes contextual representations using self-attention mechanisms. The final prediction is obtained from the [CLS] token representation:

$$\hat{y} = \text{softmax}(W\, h_cls + b)$$

This allows the model to capture contextual meaning beyond individual word occurrences.

We evaluate the following models:
- Logistic Regression (baseline)
- RoBERTa (base)

3. *Results*

We compare the performance of surface-level lexical models and deep contextual representations.

As shown in Table 2, RoBERTa consistently outperforms Logistic Regression across all evaluation metrics.

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.8536	0.8476	0.862	0.855
RoBERTa	0.9137	0.9056	0.924	0.914

Table 2: Performance comparison on the idiom detection task

Accuracy = (TP + TN) / (TP + TN + FP + FN)
Precision = TP / (TP + FP)
Recall = TP / (TP + FN)
F1 Score = 2 * (Precision * Recall) / (Precision + Recall)

The improvement is particularly noticeable in recall, indicating that RoBERTa is more effective at identifying idiomatic expressions.

Performance differences between models are consistent across validation and test splits.

As illustrated in Figure 10, transformer-based models achieve higher performance across all metrics, with the largest improvement observed in recall.

This suggests that contextual models are better suited to handling ambiguous expressions, where meaning cannot be inferred from individual words alone.

For example, the sentence: - “spill the beans about the plan”

is correctly classified as idiomatic by RoBERTa, while Logistic Regression may misclassify it due to reliance on surface-level lexical features.

Overall, the visual comparison (Figure 10) confirms that contextual representations improve model performance, particularly in challenging cases involving semantic ambiguity.

As shown in Figure 10, Model comparison on the idiom detection task (F1 score).

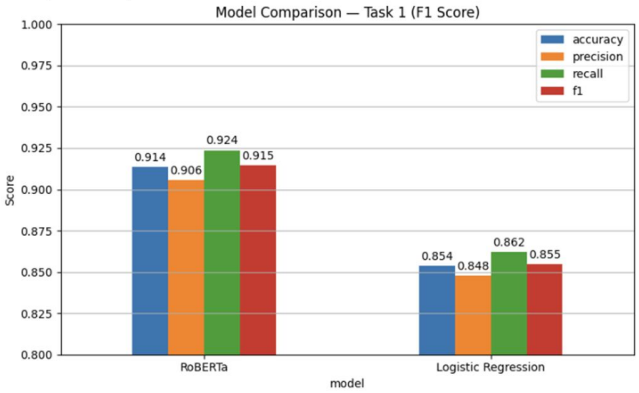

Figure 10: Model comparison As shown in Figure 10 task (F1 score)

The performance gap is visible across all metrics, with the largest improvement in recall, indicating that RoBERTa is better at correctly identifying idiomatic expressions.

The visual comparison confirms that transformer-based models provide a strong advantage in understanding context and meaning, leading to more accurate and reliable predictions

For example, transformer models correctly classify context-dependent idioms such as:
“***spill the beans***” in figurative usage, where Logistic Regression may fail due to reliance on surface patterns.

4. *Calibration and Confidence*

To further analyze model behavior, we examine confusion matrices, ROC curves, precision–recall curves, and confidence distributions.

The confusion matrices (Figure 11) show that RoBERTa reduces both false positives and false negatives compared to Logistic Regression, indicating more balanced performance across classes.

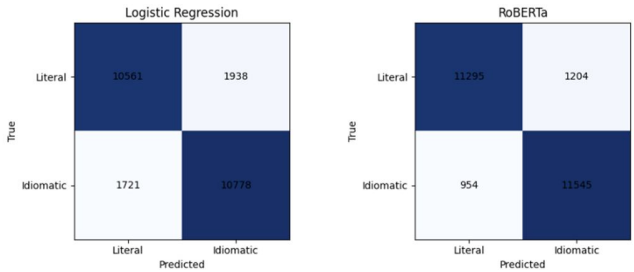

Figure 11: Confusion matrices comparison

The ROC curve (Figure 12) shows that RoBERTa achieves a higher AUC (0.973) compared to Logistic Regression (0.926), which serves as our baseline model, demonstrating stronger discriminative ability across decision thresholds.

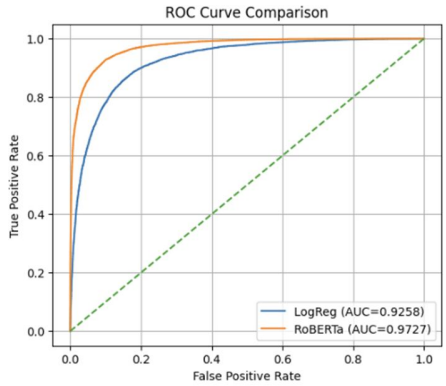

Figure 12: Roc Curve Comparison

Similarly, the precision–recall curve (Figure 13) indicates that RoBERTa maintains higher precision across recall levels, confirming its robustness in identifying idiomatic expressions.

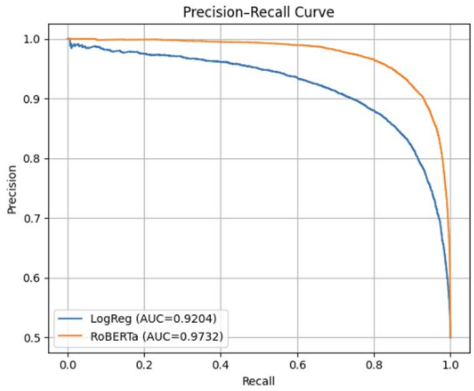

Figure 13: Precision-Recall Curve Comparison

The confidence distribution (Figure 14) shows that RoBERTa produces highly confident predictions for correct classifications, while incorrect predictions are generally associated with lower confidence.

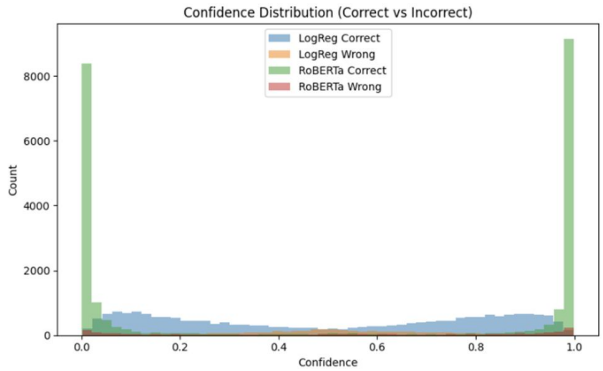

Figure 14: Confidence Distribution Comparison (Correct vs Incorrect)

For example:
- Correct prediction (high confidence):
"He finally spilled the beans about the plan" → idiomatic (confidence ≈ 0.99)
- Incorrect prediction (lower confidence):
Short or ambiguous phrases such as "baby sister" may produce lower confidence and occasional misclassification.

In contrast, Logistic Regression exhibits greater overlap between correct and incorrect confidence values, indicating less reliable probability estimates.

Overall, RoBERTa provides more meaningful and better-calibrated confidence scores.

The calibration curve (Figure 15) shows how well the predicted probabilities match the true outcomes. The dashed diagonal line represents perfect calibration.

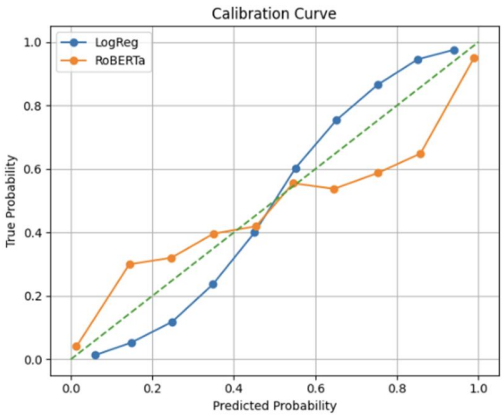

Figure 15: Calibration Curve Comparison

RoBERTa stays closer to this ideal line across most probability ranges, meaning its predicted confidence aligns better with actual correctness.

Logistic Regression deviates more from the diagonal, especially at higher confidence levels, indicating that it can be overconfident or less accurate in its probability estimates.

This result demonstrates that transformer models not only improve classification performance but also provide more trustworthy probability estimates, which is important for real-world decision-making systems.

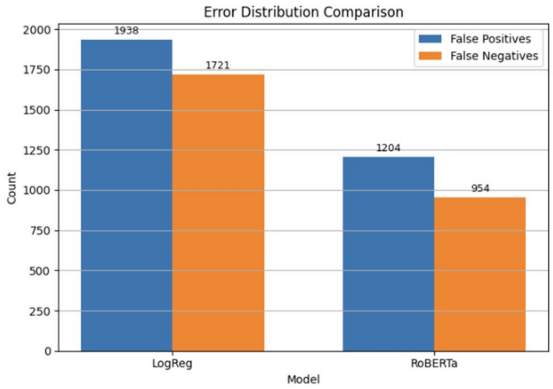

Figure 16: Error count comparison

Qualitative Example

To further illustrate model behavior, we present a representative example demonstrating RoBERTa's ability to distinguish between idiomatic and literal usage based on context.

Sentence 1:
"*He finally kicked the bucket after a long illness.*"
Prediction: **Idiomatic**.
Probability (idiomatic): 0.9973.

Sentence 2:
"*She kicked the bucket across the yard.*"
Prediction: **Literal**.
Probability (idiomatic): 0.0066.

This example highlights that the model correctly identifies the figurative meaning in the first sentence while recognizing the literal action in the second. The high confidence in both predictions indicates strong contextual understanding and reliable decision-making.

This demonstrates that the model is able to rely on contextual cues rather than surface-level word matching, correctly distinguishing between identical phrases used in different meanings.

5. *Error Analysis and Key Failure Modes*

(1) Short or context-limited inputs
Short phrases such as "baby sister" or "Cuban bread" lack sufficient context, making it difficult to determine whether the usage is literal or idiomatic.

(2) Named entities and domain-specific expressions
Phrases such as "Benjamin Button" or "Bravo Zulu" may be misclassified due to ambiguity between literal and idiomatic interpretation.

(3) Context-dependent idioms
Expressions like "one for the road" require broader discourse context, which may not be fully captured in single-sentence inputs.

(4) Ambiguous phrases
Certain expressions can be interpreted both literally and idiomatically depending on context, representing inherent challenges in the task.

Table 3 presents representative examples of false negatives (missed idioms) and false positives (incorrect idiom predictions).

Sentence	True Label	Idiomatic Probability
The hikers waited out the night in the tent during the heavy rainfall.	Idiomatic	0.017
After fixing the machine quickly, he could have a good time of it for the rest of the day.	Idiomatic	0.056
The soldiers were taught to duck down quickly to evade enemy fire.	Idiomatic	0.08
I won't wear new shoes tomorrow because my dogs are barking today.	Idiomatic	0.016
He said he would help, but only when two Sundays meet.	Idiomatic	0.476

Table 3: Examples of false negatives.

Table 4 Examples of false positives (literal sentences misclassified as idiomatic).

Sentence	True Label	Idiomatic Probability
Close their eyes and think of England.	Literal	0.527
The pairing of the spicy chili and the salty cheese was a match made in hell for my sensitive stomach.	Literal	0.692
She chose to take his words at face value, believing every promise without looking for hidden meanings.	Literal	0.998
A true friend of the environment would take steps to reduce waste and pollution.	Literal	0.928
He needed some solitude to rest his soul after hearing the sad news.	Literal	0.766

Table 4: Examples of false Positives.

These examples show that most errors occur in sentences where literal and idiomatic meanings overlap or where contextual cues are limited.

False negatives typically involve idioms expressed in subtle or less explicit forms, while false positives often arise when literal sentences contain words commonly associated with idiomatic expressions.

This highlights that remaining errors are primarily due to semantic ambiguity rather than simple lexical confusion.

6. *Summary*

Overall, the results suggest that idiom detection benefits from contextual understanding.

While Logistic Regression provides a strong baseline, its reliance on surface lexical features limits its ability to handle ambiguous expressions. In contrast, RoBERTa achieves higher accuracy and more balanced error distribution by leveraging contextual representations.

However, remaining errors highlight the inherent ambiguity of idiomatic language and the importance of richer contextual and semantic understanding in future work.

B. *Task 2: Context-to-Idiom Generation*

In this task, the model is given a contextual sentence or semantic description and is required to predict the most appropriate idiomatic expression. Unlike idiom detection, this task requires both semantic understanding and generation capability, making it more challenging and closer to real-world language use.

Formally: $y^* = \text{argmax}_y P(y \mid x)$
where y is selected from a predefined idiom inventory.

A. *SBERT Retrieval Model*

Each context x and candidate idiom i are encoded into vector representations:

$h_x = \text{Encoder}(x)$, $h_i = \text{Encoder}(i)$

Similarity is computed using cosine similarity:
$\text{sim}(x, i) = (h_x \cdot h_i) / (\|h_x\| \, \|h_i\|)$

Prediction: $\hat{y} = \text{argmax}_i \, \text{sim}(x, i)$

B. *FLAN-T5 Generative Model*

The model generates an idiom sequence y autoregressively:

$P(y \mid x) = \prod P(y_t \mid y_{<t}, x)$

Prediction: $\hat{y} = \text{argmax}_y P(y \mid x)$

C. *Proposed FLAN-First Hybrid Model*

"$\hat{y}$ equals y_gen if Match $\geq \tau$, otherwise the best retrieval candidate."

$\text{Match}(y_{gen}, I) = \max_i \text{similarity}(y_{gen}, i)$

1. *Experimental setup*

$$\text{Top-K Accuracy} = (1/N) \sum \mathbb{1}\{y \in \text{TopK}(x)\}$$

$$\text{MRR} = (1/N) \sum (1 / \text{rank}_i)$$

2. *Quantitative Results:*

Model	Exact Match (Top-1)	Top-3	Top-5	MRR@100
SBERT Retrieval	0.461	0.599	0.65	0.549
FLAN-T5	0.678	-	-	-
Hybrid FLAN-first	0.818	-	-	-

Table 5: Models exact Match - Top-1, Top-2, Top-3.

Performance differences between models are consistent across validation splits.

The results show that:

SBERT provides strong semantic retrieval but suffers from ranking limitations.

FLAN-T5 improves contextual reasoning but may produce invalid or noisy outputs.

The hybrid model combines generative reasoning with retrieval grounding, improving both accuracy and robustness.

This confirms that idiom prediction benefits from a two-stage reasoning process, where generation captures semantic intent and retrieval enforces discrete correctness.

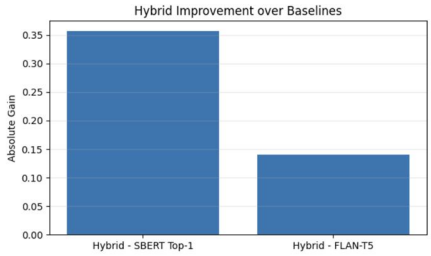

Figure 9: Hybrid Improvement over Baselines

Comparison	Gain
Hybrid vs SBERT	0.356
Hybrid vs FLAN	0.14

Table 6: comparison table gain from hybrid to SBERT and FLAN.

The hybrid model improves exact-match accuracy by 35.7 percentage points over SBERT Top-1 and by 14.0 points over FLAN-T5.

Decision	Count	Percent
FLAN exact bank	7885	0.724458
FLAN fuzzy correction	608	0.055862
SBERT fallback	2391	0.21968

Table 7: comparison table hybrid Decision Distribution.

3. *Hybrid Decision Analysis*

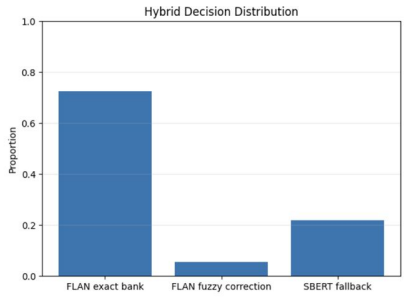

Figure 10: Hybrid Decision Distribution

Most predictions are resolved by the generative model, while retrieval provides corrective support.

4. *Error Analysis*

Errors are categorized into:

1. Retrieval Error: correct idiom not retrieved
2. Ranking Error: correct idiom not ranked first
3. Generation Error: incorrect or noisy output

The hybrid model reduces both ranking and generation errors

Model Interaction Analysis

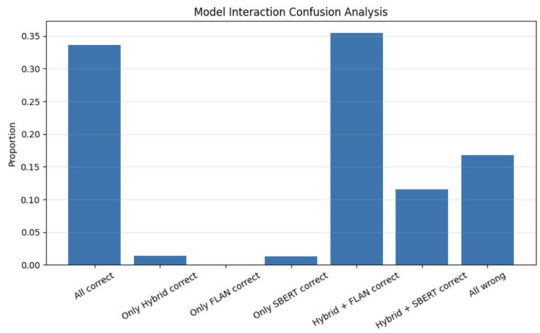

Figure 11: Model Interaction Confusion chart

These results indicate that the hybrid model is not merely averaging predictions but recovering additional correct cases through complementary decision behavior.

5. *Qualitative Analysis*

Case Study 1: Hybrid corrects both models

- SBERT retrieval fails to select the correct idiom
- FLAN generates an incorrect or approximate phrase
- The hybrid model successfully corrects the output

Case Study 2: FLAN succeeds alone

- FLAN correctly generates the idiom directly
- SBERT retrieval may rank incorrect candidates
- The hybrid model accepts the FLAN output without modification

Case Study 3: SBERT fallback saves FLAN

- FLAN produces incorrect or incomplete outputs
- SBERT retrieval successfully identifies the correct idiom
- The hybrid model switches to retrieval-based prediction

These examples demonstrate complementary strengths between generation and retrieval.

6. *Discussion*

The results indicate that idiom prediction is not purely a retrieval task, but benefits from generative reasoning.

The hybrid model effectively integrates both paradigms, improving robustness and accuracy.

The hybrid approach introduces additional computational cost due to dual processing, but this is justified by the performance gain.

This improvement arises because generative models capture semantic flexibility, while retrieval models enforce discrete correctness within a constrained idiom inventory

7. *Task Summary*

The proposed hybrid model achieves the best performance on the context-to-idiom task.

By combining generative reasoning with retrieval-based grounding, it overcomes the limitations of both approaches and provides a robust solution for idiomatic understanding.

C. *Task 3: Arabic Context-to-Idiom (Retrieval Benchmark)*

In Task 3, the objective is to retrieve the canonical English idiom corresponding to an Arabic contextual sentence. Unlike Task 2, which is framed as a generation task, Task 3 is formulated as a cross-lingual semantic retrieval problem. The input is the Arabic idiomatic context, and the target is the associated canonical English idiom from a predefined inventory.

The task is evaluated as a ranking problem, where the correct idiom must be retrieved from a predefined candidate set.

We formulate the Arabic context-to-idiom task as a semantic retrieval problem rather than a generative task. This design choice is motivated by the fixed and canonical nature of idioms, where exact canonical matching is required at evaluation time. Retrieval-based approaches allow mapping Arabic contextual meaning into a structured embedding space and retrieving the most semantically similar idiom from a fixed candidate set.

1. *Formulation*

Each Arabic query x and candidate idiom i are mapped into vector representations:

$h_x = \text{Encoder}(x)$, $h_i = \text{Encoder}(i)$

Similarity is computed using cosine similarity:

$$sim(x, i) = (h_x \cdot h_i) / (\|h_x\| \|h_i\|)$$

Prediction is defined as:

$$\hat{y} = \text{argmax}_i \, sim(x, i)$$

where sim(·) denotes cosine similarity in a shared embedding space.

2. *Experimental Setup*

To construct the Task 3 benchmark, Task 3 was initially explored through several generation-based, hybrid, and retrieval-based scripts, including mT5 baselines, hybrid mT5–retrieval variants, and Arabic-bank retrieval models. Early experiments used a broad Arabic input formulation that combined multiple Arabic fields and adopted a strict split strategy designed to avoid overlapping at the idiom or canonical target level. While this setup was methodologically strict, it produced **near-zero retrieval accuracy**, indicating that the benchmark was too difficult in its initial form and poorly matched to the intended retrieval setting.

To address this, the task was reformulated as an example-level cross-lingual retrieval problem.

The final benchmark retained only idiomatic examples, used `idiom_in_example_arabic` as the Arabic query, and used `idiom_canonical` as the English retrieval target.

Instead of splitting by idiom identity, the data was split by example within each idiom, allowing the same idiom to appear in both train and test through different contextual realizations.

This setup evaluates whether the model can retrieve the correct idiom for a novel Arabic context while preserving a realistic candidate inventory.

Under this revised formulation, the final retrieval pipeline showed a clear progression with fine-tuned E5 (v2) where Top-1 = 64.86%, Top-5 = 81.46%

These results show that stronger pretrained multilingual embeddings provide only modest gains by themselves, whereas task-specific contrastive fine-tuning is the dominant factor behind performance improvement. Additional gains are obtained through richer supervision, such as using multiple positives per query and longer training. The final best-performing Task 3 system is therefore the upgraded fine-tuned multilingual E5 retrieval model.

3. *Arabic Semantic Bank Construction*

A semantic bank is constructed from the training split, where each idiom is represented through multiple Arabic textual views:

- Example sentences (idiom usage)
- Example meanings
- Canonical meanings

Each entry is encoded into a multilingual embedding space, resulting in a dense representation matrix. This multi-view representation improves semantic coverage and allows the model to match queries against diverse contextual signals.

4. *Models Evaluated*

We evaluate a range of multilingual retrieval models, including both baseline and fine-tuned configurations:

- Multilingual MiniLM (baseline)
- Multilingual E5 (base, zero-shot)
- Multilingual E5 (fine-tuned v1)
- Multilingual E5 (fine-tuned v2 with enhanced positives)

While all models are evaluated, detailed analysis focuses on the strongest-performing configurations.

5. *Quantitative Results*

Evaluation is conducted using Top-K accuracy metrics:

$$\text{Top-K Accuracy} = (1/N) \sum \mathbb{1}\{y \in \text{TopK}(x)\}$$

Model	Top-1	Top-5
Zero-shot MiniLM baseline	23.27%	31.79%
Zero-shot multilingual E5	28.16%	39.08%
Fine-tuned E5 v1	60.37%	76.07%
Fine-tuned E5 v2	64.86%	81.46%

Table 8: Models Accuracy comparison table.

The multilingual E5 model consistently outperformed the MiniLM baseline. Top-1 accuracy improved from 23.27% to 27.46%, while Top-5 accuracy increased from 31.79% to 38.34%. These gains indicate that instruction-aware multilingual retrieval embeddings provide stronger semantic alignment between Arabic contextual inputs and canonical English idiom candidates.

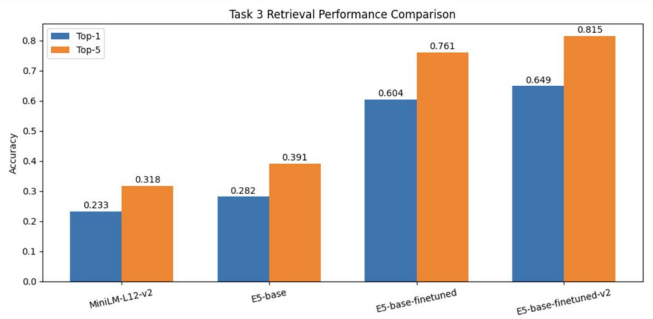

Figure 12: Model Retrieval Performance Comparison.

The results show a substantial improvement from fine-tuning, with E5-base-finetuned v2 outperforming all other models.

6. *Recall Analysis*

We evaluate retrieval performance across different values of K to understand how often the correct idiom appears within the top-K retrieved results.

The recall curve shows that:

- Recall increases rapidly for small values of K.
- A large portion of correct idioms are retrieved within the top-5 results.

- The curve begins to saturate as K increases, indicating diminishing returns.

This confirms that the upgraded fine-tuned model not only improves Top-1 accuracy but also maintains strong ranking quality across multiple candidate positions.

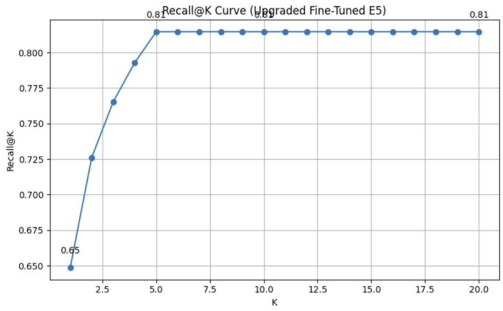

Figure 13: Recall curve (upgraded fine-tuned E5 v2)

7. *Confidence and Similarity Analysis*

To better understand model behavior, we analyze the relationship between similarity-based confidence scores and prediction accuracy.

Results show a strong positive correlation between confidence and correctness. Predictions with higher cosine similarity scores are significantly more likely to be correct.

Accuracy increases steadily across confidence bins, reaching over 90% accuracy in the highest confidence range.

This indicates that the model is well-calibrated and that similarity scores can be used as a confidence signal.

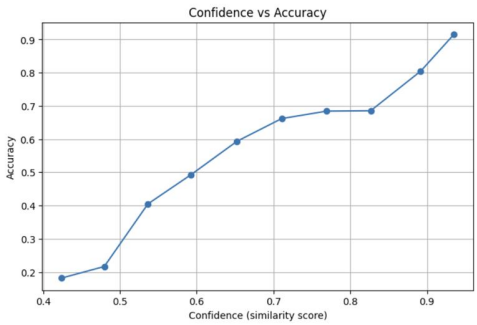

Figure 14: Confidence vs accuracy across similarity bins

The similarity score distribution shows a clear separation between correct and incorrect predictions:

- Correct predictions tend to have higher similarity scores.
- Incorrect predictions are concentrated in lower score regions.
- However, there is some overlap, indicating difficult cases.

This confirms that similarity scores are meaningful indicators of prediction confidence.

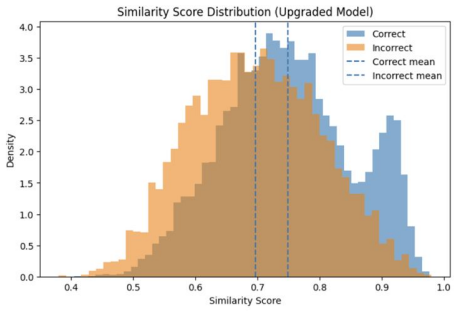

Figure 15: Distribution of cosine similarity scores

8. *Score Distribution Comparison*

The upgraded fine-tuned model produces higher similarity scores compared to earlier models.

This indicates improved embedding alignment, stronger separation between correct and incorrect matches, better ranking confidence.

The shift in distribution confirms the effectiveness of fine-tuning.

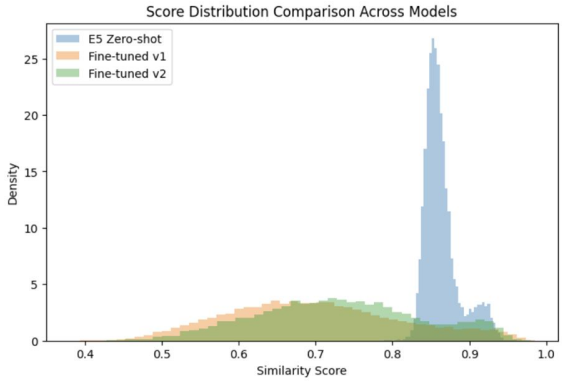

Figure 16: Score Distribution Comparison

9. *Abalation study*

The ablation results show that:

- Replacing MiniLM with zero-shot E5 yields a modest gain.
- Fine-tuning produces the largest improvement by far.
- The upgraded training strategy provides an additional performance boost.

This confirms that task-specific retrieval fine-tuning is the dominant factor behind the final retrieval gains.

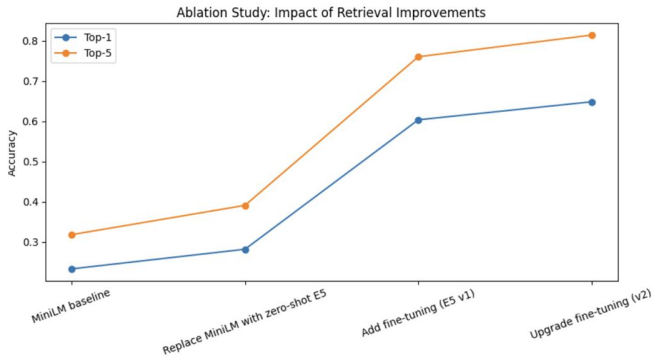

Figure 17: Ablation study: Impact of Retrieval Improvement

10. *Error Analysis*

Error analysis reveals several key failure modes:

(1) Semantic mismatch:
Some Arabic inputs describe concepts that do not map cleanly to a single English idiom, resulting in approximate predictions.

(2) Surface similarity bias:
The model sometimes favors idioms with overlapping lexical patterns rather than true semantic alignment.

(3) Ranking limitations:
In many cases, the correct idiom appears within the Top-5 predictions but is not ranked first, indicating limitations in fine-grained ranking.

(4) Cross-lingual ambiguity:
Certain Arabic expressions have multiple valid interpretations, making it difficult to map to a single idiomatic equivalent.

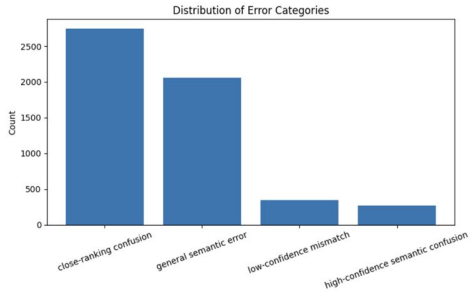

Figure 18: Error distribution across prediction

These findings suggest that while retrieval models capture general semantic similarity, they may struggle with precise idiomatic alignment, particularly in cross-lingual settings.

11. *Qualitative Cross-Lingual Examples*

Qualitative examples demonstrate that the model is able to retrieve semantically appropriate idioms in many cases. For instance, Arabic expressions describing inevitability or shared emotional burden are correctly mapped to idioms such as “come to an end” and “a problem shared is a problem halved.”

However, errors often occur when the input is ambiguous or when multiple idioms share similar meanings, highlighting the difficulty of selecting a single correct expression.

To illustrate model behavior, we present representative examples of Arabic input queries and their retrieved idioms.

Example 1:
Arabic: "انتهى كل شيء في النهاية"
Prediction: "come to an end"
This example demonstrates correct semantic alignment between the Arabic description of inevitability and the corresponding idiomatic expression.

Example 2:
Arabic: "شارك همومك مع الآخرين لتخفيفها"
Prediction: "a problem shared is a problem halved"
The model successfully captures the underlying meaning of shared emotional burden.

Example 3:
Arabic: "لا توجد أي فرصة لحدوث ذلك"
Prediction: "not a chance"
This example shows accurate mapping of strong negation to an idiomatic expression.

Example 4 (Error Case):
Arabic: "يجب أن تبطئ كلامك حتى يفهمك الجميع"
Prediction: "please speak more slowly"
Target: "slow down"
Here, the model retrieves a semantically related phrase rather than the canonical idiom, illustrating surface similarity bias.

12. *Discussion*

The results indicate that Arabic context-to-idiom retrieval is substantially more challenging than its monolingual counterpart.

Unlike direct generation tasks, retrieval-based approaches rely on the quality of the embedding space to bridge semantic gaps between languages. While multilingual models improve alignment, they are still sensitive to subtle semantic differences and cultural variations in idiomatic usage.

Overall, this task highlights the limitations of purely retrieval-based approaches in cross-lingual idiom understanding and motivates the need for hybrid or generative methods that can better capture semantic nuance.

Replacing the MiniLM baseline with multilingual E5 improved retrieval quality substantially. Top-1 accuracy increased from 23.27% to 27.46%, while Top-5 accuracy rose from 31.79% to 38.34%. This suggests that instruction-aware multilingual retrieval embeddings provide better alignment between Arabic contextual queries and canonical English idioms.

13. *Task Summary*

The multilingual E5 model provides the strongest performance for cross-lingual idiom retrieval, outperforming SBERT across all metrics.

However, the moderate accuracy levels and observed error patterns indicate that idiom understanding across languages remains an open challenge, requiring more advanced modeling strategies beyond similarity-based retrieval.

VI. Discussion

The experimental results demonstrate that IdiomX supports a wide range of NLP tasks with meaningful performance.

The experimental findings highlight three key insights. First, idiom detection requires contextual modeling rather than surface-level lexical matching. Second, context-to-idiom generation is not purely a retrieval problem, but requires semantic reasoning. Third, hybrid approaches outperform individual paradigms by combining generative flexibility with retrieval-based grounding.

The experiments suggest that idiom understanding benefits from a division of labor: generative models provide flexible contextual reasoning, while retrieval mechanisms provide discrete grounding and correction.

VII. Applications

The IdiomX dataset supports a wide range of natural language processing tasks, including:

- idiom detection
- idiom generation
- cross-lingual idiom mapping
- multilingual figurative language evaluation

The dataset is particularly valuable for evaluating transformer-based models and large language models in handling non-compositional language.

VIII. Dataset Availability

The IdiomX dataset is publicly available via GitHub and Hugging Face to support reproducibility and further research. The repository includes the dataset, preprocessing scripts, and documentation for easy integration

IX. Limitations

While IdiomX provides large-scale coverage and rich annotations, part of the dataset is generated using LLM-based methods, which may introduce minor inconsistencies or stylistic biases. Arabic translations primarily reflect standard written Arabic and may not fully capture dialectal variations.

Future work includes incorporating human annotation and extending the dataset to additional languages.

Some generated idiomatic or literal examples may reflect model-preferred phrasing rather than naturally occurring corpus frequency.

Additionally, evaluation for the generation task relies primarily on exact match metrics, which may not fully capture semantic equivalence between idioms.

X. Conclusion

In this work, we introduced IdiomX, a large-scale bilingual dataset for idiomatic expression understanding, providing extensive coverage, rich semantic annotations, and near-complete English–Arabic alignment.

Through comprehensive benchmarking, we demonstrated that idiom understanding requires deep contextual modeling. Transformer-based models significantly outperform lexical approaches on idiom detection, while hybrid generation–retrieval methods achieve the best performance on context-to-idiom prediction.

For the cross-lingual Arabic context-to-idiom task, we showed that retrieval-based models can capture general semantic alignment but remain limited in precise idiomatic mapping. Fine-tuning substantially improves performance, yet challenges such as semantic ambiguity, ranking limitations, and cross-lingual variation persist.

These results highlight that idiom understanding is inherently complex and cannot be fully addressed by a single modeling paradigm. Future work should explore hybrid architectures that integrate retrieval, generation, and cross-lingual reasoning to better capture idiomatic meaning across languages.

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